NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES

Marks: 20	**QUIZ-3**	**Time: 20 Minutes**

Subject: Probability and Statistics **Section: G**
Instructor: Hazber Samson Date: 17-03-2026
Student Name_____ Roll No_____

Q1. **(4+4+4+4+4=20)**

Consider the following joint probability density function of the random variables *X* and *Y* :

$$f(x,y)=\begin{cases} k(3x-y) & 1<x<3,\ \ 1<y<2 \\ 0 & elsewhere \end{cases}$$

(a) Find the value of k then find the marginal distributions functions of *X* and Y.
(b) Find $P(1<X<2, 1<Y<2)$
(c) Check whether X and Y are independent or not.
(d) Find $E(XY),\ E(X),\ E(Y)$
(e) Find $COV(X,Y)$ *and* $Corr(X,Y)$

GOOD LUCK !